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Molecularly imprinted poly(meta-phenylenediamine) based QCM sensor for detecting Amoxicillin

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ABSTRACT

A chemical sensor based on molecularly imprinted polymer (MIP) and quartz crystal microbalance (QCM) to detect amoxicillin (AMO) antibiotics in aqueous samples was developed. The thin film of AMO-MIP was generated electrochemically from meta-phenylenediamine (mPD) directly on the QCM transducer. Pre-polymerization complex formation between the template (AMO) and the monomer molecules (mPD) was confirmed by a combination of computational modeling and spectroscopic studies. The electrodeposition process was carefully studied to allow for the selection of the optimal parameters for stable AMO-MIP film deposition. The AMO-MIP QCM sensor showed a significantly better sensitivity and affinity than the reference film displaying more than seven times relative adsorption capacity and a limit of detection down to 0.2 nM. Likewise, the sensor demonstrates good selectivity to the target analyte (AMO) than the other non-templated molecules and remain sensitive to the target even after a prior exposure to other interferents that may be present within the same environment. This remarkable result in the analysis of amoxicillin on QCM sensor without employing any signal amplification methodology demonstrates an important step towards the fabrication of MIP-based environmental sensor.

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1. Introduction

The environmental impacts of antibiotics pollution cannot be over-emphasized due to their potential of increasing bacterial resistance [1,2]. Their urgent detection, especially in aqueous environment and food has therefore continued to attract enormous research interests. As a result, analytical devices which combine the selectivity of biological systems with the chemical stability of chemical systems are increasingly becoming a focus of attention in environmental and clinical research [3]. Molecularly imprinted polymers (MIPs) possessing pre-defined cavities for specific recognition of target analytes, are artificial receptors with potential for fabricating state-of-the-art analytical devices. Specific cavities of MIPs are pre-formed by the polymerization of suitable functional monomers in the presence of the target molecules, acting as templates, which after subsequent removal reveal complementary sites capable of selectively recognizing the target molecule by size, shape and functional groups. Antibiotics constitute a larger share of pharmaceuticals contributing to the rising pollution of the aquatic environment [4,5]. Amoxicillin is a broad-spectrum antibiotic that belongs to one of the most widely known antibiotic class, the penicillins. It has low metabolic rate in humans and as a result, up to 90% is being excreted in the raw form [6] and therefore exists in various water bodies in the concentration range of 0.18–4.57 nM [7]. Since amoxicillin possesses a variety of functional groups potentially capable of interacting with functional monomers through the formation of non-covalent bonds, it has the tendency to create recognition cavities that are complementary to those of the template molecules in the resulting imprinted polymer following template extraction.

Thin films of polymers are the most suitable for sensing in label-free configuration [8]. Polymers generated from functional monomers consisting of phenylenediamine parent units are gaining wide popularity among researchers due to their beneficial electroactivity and suitability for a wide variety of applications, including energy storage, electrocatalysis, electrochromism, and sensors [9–12]. Their appropriateness as polymer matrix in molecular imprinting stems from the fact that the growth of their polymer films results in a rigid structure demonstrating important recognition functionalities relevant for molecular recognition, especially by non-covalent interactions [13]. More importantly, the possibility of generating a thin but continuous polymer film that

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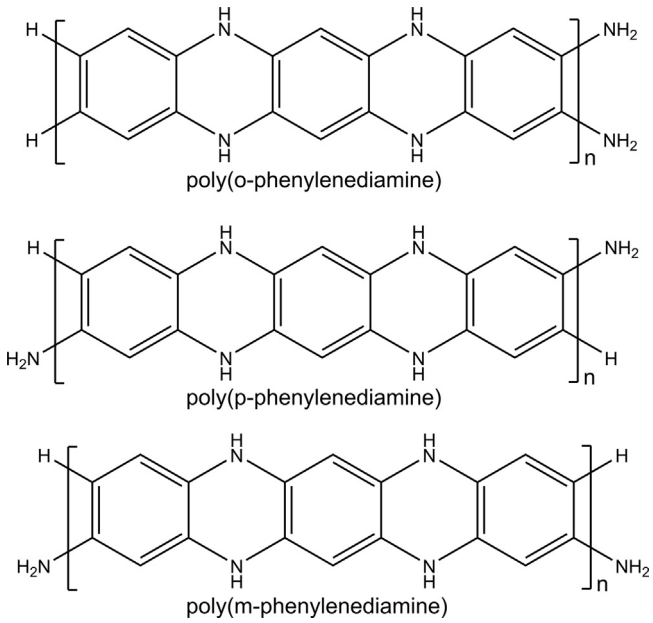

Fig. 1. Ladder-like structures for the polymers of the three isomers of phenylenediamines; oPD, pPD and mPD.

eliminates the long diffusion path through which analytes have to travel to the pre-formed active sites makes them quite preferable for fabricating molecular imprinting chemosensor [14,15]. Consequently, many sensors are being fabricated on piezoelectric and other transducer surfaces via electrochemically synthesized films of poly(phenylenediamine) leading to extended linear concentration range and analytically significant limit of detection (LoD) [16,17]. Of the three isomers of poly(phenylenediamine); Poly(*o*-phenylenediamine), P(oPD), Poly(*m*-phenylenediamine), P(mPD), and Poly(*p*-phenylenediamine), P(pPD) (see Fig. 1), P(oPD)'s structure has been substantially defined while the structure of P(mPD) has not been fully elucidated because the P(mPD) obtained by chemical or electrochemical oxidative polymerization is almost insoluble in most solvents [18]. Nevertheless, FTIR spectroscopic studies [19] demonstrated no significant difference in the chemically and electrochemically synthesized polymers of these isomers in the IR spectra. These polymers have been estimated to yield ladder-like structures [20].

Also, according to the report of Rhemrev-Boom et al. [21], the structural information obtained from electrosynthesized P(mPD) and P(oPD) films by means of ATR-FTIR spectroscopy permitted the conclusion that the formation of P(mPD) films was comparable to that of P(oPD) films. Furthermore, Killoran and O'Neill [22] studied the electrosynthesis and permselective properties of the three poly(phenylenediamine) films and concluded that P(mPD) polymerized faster than P(oPD) and P(pPD) films under identical polymerization conditions and it is the least soluble in methanol and other solvents. These properties of P(mPD) allow its higher suitability for MIP electrosynthesis and its subsequent regeneration by hash solvents such as methanol mixture of acetic acid. Consequently, we have recently reported the detection of small molecule (e.g. sulfamethizole) and large molecule (e.g. IgG) by using an electropolymerized p(mPD) film on a SAW sensor transducer [23,24]. The resulting sensors displayed interesting features including high selectivity and satisfactory stability in aqueous media thus, further establishing the important use of P(mPD) as matrix in molecular imprinting.

Although, MIPs for antibiotic detection have seen quite reasonable publications however, MIP chemosensors exhibiting amoxicillin detection in aqueous media using thin P(mPD) films integrated with QCM sensors and without the use of signal amplifying nanomaterials have not been reported. In this report, an amoxicillin MIP film was developed using electrosynthesized P(mPD) polymer matrix integrated with a quartz crystal microbalance (QCM) sensor having a gold coating as the sensor transducer to facilitate a direct communication of the imprinted polymer film with the transducer during the polymerization and rebinding analysis. To prepare MIPs with a high density of imprinted sites [16], we have investigated different electropolymerization conditions for the synthesis of the MIP films to optimize the binding capacity of the resulting antibiotic-imprinted polymeric material (AMO-MIP) and improve the performance of the fabricated chemosensor. Important parameters of the developed sensor were studied and discussed.

2. Experimental

2.1. Materials and methods

Acetic acid, sulfuric acid and hydrogen peroxide were provided by Lach-ner (Czech Republic). The other chemicals were obtained from Sigma-Aldrich. All chemicals were of analytical grade or higher and were used as received without any further purification. Ultrapure water (resistivity 18.2 MΩ cm, Millipore, USA) was used for preparation of all aqueous solutions. Phosphate buffered saline (PBS) solution (0.01 M, pH 7.4) was used to prepare synthesis and analyte solutions. The gold (Au) electrodes of a 5 MHz QCM (Maxtek, Inc.) was used in this work for antibiotic-MIP and the reference non-imprinted polymer (NIP) film deposition. $Ag/AgCl/KCl_{sat}$ electrode was used as a reference electrode in all electrochemical measurements. Before electrochemical deposition of the films, the QCM sensors were cleaned for 3 minutes in hot piranha solution consisting of 30% H_2O_2 and concentrated H_2SO_4 in 1:3 ratio followed by electrochemical cleaning in 0.1 M H_2SO_4 aqueous solution by cycling the electrode potential in the range of −0.2 to 1.5 V with a scan rate of 50 mV/s, until the cyclic voltammograms were reproducible. This was followed by thorough washing with ultrapure water and drying in nitrogen stream. Electrochemical measurements were made in a custom-made 25 mL Teflon electrochemical cell connected with Reference 600 Potentiostat (Gamry Instruments, USA). A conventional three-electrode system was used with QCM sensor electrode as the working electrode, a rectangular shaped platinum plate ($4 \times 1.5\ cm^2$) as a counter electrode, and $Ag/AgCl/KCl_{sat}$ as a reference electrode.

2.2. Polymer matrix optimization

2.2.1. Computer modeling

The mPD complex formed with the antibiotic template was generated by GaussView 5.0.9 software. Its geometry was also optimized by semi-empirical PM3 method with Gaussian'09 software. The conformational optimizations and binding energies for the complex were calculated by density functional theory (DFT) method at B3LYP/6–31 + G level with Gaussian'09 software.

2.2.2. Spectroscopic analysis

UV–vis absorption spectra of the PBS buffer solutions containing individual components, mPD and AMO as well as their mixture were recorded in a 1 cm quartz cuvette with the Shimadzu 2401 spectrophotometer in the wavelength range from 200 to 320 nm. The changes in absorbance of the solution mixtures are then compared with the spectra of the individual components as references. ^{13}C NMR spectroscopy measurement was conducted using Bruker SMART X2S benchtop diffractometer model. The samples were prepared in D_2O solution as solvent. The molar ratio of template to monomer (AMO: mPD) in the solution mixture was kept at 1:1.

2.3. Amoxicillin-MIP preparation and characterization

2.3.1. Electrosynthesis of antibiotic-MIP films

Thin films of P(mPD) and/or P(mPD)/AMO were electrodeposited on QCM sensors. This was achieved by applying a constant potential (0.6 V vs Ag/AgCl/KCl_{sat}), in PBS buffer solution containing 5 mM mPD and/or mixture of 5 mM mPD and 5 mM AMO, keeping the monomer and template concentration ratio at 1. To obtain similar film thickness, the mPD electrochemical polymerization in the presence of the antibiotics was monitored by the electrochemical quartz crystal microbalance (EQCM). The EQCM measurements were performed using the QCM100 system (Stanford Research Systems, Inc., Sunnyvale, CA, USA) connected to the reference 600TM potentiostat (Gamry Instruments, Inc.) and the PM 6680 B counter (Fluke Corporation) as previously described elsewhere [25]. Film syntheses were continued until the resonant frequency dropped to the designated value (−400 Hz). As a reference, non-imprinted polymer (NIP) was also synthesized at the same conditions but without adding any antibiotics in the synthesis solution. The deposited polymer films were observed to be uniform. The films' thicknesses were then estimated by converting the frequency shift to the corresponding mass by applying the Sauerbrey's equation (Eq. (1)) and dividing the latter by the polymer's density (Eq. (2)). After polymer film electrochemical deposition, the electrode was rinsed with distilled water and dried in a nitrogen stream.

$$\Delta f = -f_0{}^2 \Delta \mathrm{m}/\mathrm{N}\rho_\mathrm{q} = -C_f \Delta \mathrm{m} \quad (1)$$

$$t_f = \Delta m/\rho_f \quad (2)$$

where Δf – resonant frequency shift (Hz), f_0–fundamental frequency of the crystal (Hz), Δm – mass change (g/cm^2), N – frequency constant for quartz (167 kHz cm), ρ_q – density of quartz (2.65 g/cm^3), C_f – sensitivity factor (for 5 MHz quartz crystal, 56.6 Hz mg^{-1} cm^2), t_f – thickness of film, ρ_f – density of film.

2.3.2. Characterization of the prepared films

After electrochemical deposition of P(mPD)/AMO film, the removal of the antibiotic molecules from the polymer matrix was carried out to create AMO-MIP film. For this purpose, the polymer modified electrode was immersed in acetic acid/methanol (1:1) solution and held there for 24 h, then the eluted solution was analyzed by UV–vis absorption spectroscopy and amoxicillin concentration was determined at the specific wavelength, λ= 274 nm, of its absorbance. The concentration in the washing solution was also determined by chromatographic measurements using a high-performance liquid chromatography combined with diode array detector and mass-spectrometer, (HPLC-PDA-MS, Shimadzu LC–MS 2020). Phenomenex Gemini-NX 5 u C18 110A 150 × 2.0 mm column, inner diameter 1.7 μm, was used with two eluents; 0.1% acetic acid aqueous solution (eluent A), and acetonitrile (eluent B), with total eluents flow of 0.3 mL min^{-1}. The antibiotic amount in eluted solution was determined and compared with theoretically estimated amount of the antibiotic presented in the polymer film.

2.3.3. Rebinding study

Rebinding of the target AMO molecules on the prepared AMO-MIP film was studied by QCM-FIA technique, which allows real-time monitoring of molecular interactions on the surface of the QCM sensor. The rebinding analysis was carried out in the QCM-FIA system comprising one programmable precision syringe pumps (Cavro XLP 6000® XLP 6000, Tecan Nordic AB, Mölndal, Sweden), a motorized six-way port injection valve (C22-3186EH, VICI® Valco Instruments Company Inc., USA) controlled by a microelectric actuator and a small volume (150 μL) axial flow cell attached to the QCM sensor holder (Stanford Research Systems, Inc.). Sample injection was carried out by a 5 mL disposable plastic syringe. All elements of the system were connected to a PC and controlled by software written in Labview. A constant flow of degassed PBS buffer solution (pH = 7.4) flowed over the sensor at a flow rate of 25 μL/min until a constant baseline of the QCM sensor resonance frequency was reached. Subsequently, the increasing concentrations of the analyte samples (1.6 μM, 8 μM, 40 μM, 200 μM, 1 mM of the AMO in PBS buffer) were injected into the flow stream via an injection loop (500 μL). Prior to the rebinding study, the P(mPD)/AMO films as well as their P(mPD) references were regenerated by immersing the QCM sensor modified films in acetic acid/methanol (1:1) solution for 24 h. The sensorgrams recorded were then fitted to the appropriate binding model to determine the equilibrium responses for both the MIPs and NIPs surfaces. These equilibrium data were then used for the construction of the binding isotherms.

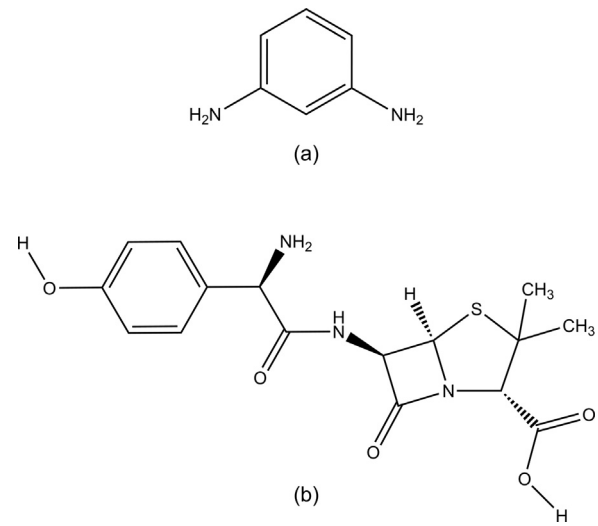

Fig. 2. Structural formulae of the (a) functional monomer, mPD and (b) target molecule, AMO used for the MIP formation.

3. Results and discussion

3.1. Polymer matrix optimization

In a non-covalent or self-assembly molecular imprinting, selectivity of a MIP depends, to a great extent, on the nature and strength of interaction of pre-polymerization complex formed between the target analyte and functional monomers in a porogenic solvent [26]. Therefore, a preliminary study of molecular complexes between a monomer and an analyte (template) molecule is required to design a MIP with optimal performance. This can be realized by a combination of both theoretical (computational modeling) and experimental (spectroscopic analysis) approaches [27].

3.1.1. Computational modeling

Geometrically optimized structure of the pre-polymerization complex formed between the mPD monomer and the AMO antibiotic template molecules (Fig. 2) was estimated by density functional theory (DFT) calculations using Gaussian'09. The strength of interaction was analyzed in terms of the binding energy, length of hydrogen bonds and Mulliken charges. To appreciate the binding energy value, the binding energy of complexes formed between AMO and other electropolymerizable monomers; pyrrole (Py) and 3,4-ethylenedioxythiophene (EDOT) were also estimated by DFT calculations. Table 1 shows the total binding energy as well as the average hydrogen bond length of the monomer-template complexes. The sites of interaction, binding energy for each site, lengths

Table 1
Binding energies of the complexes formed between AMO and three different monomers, mPD, EDOT and Py as calculated by Gaussian'09.

Template-monomer complex	Binding energy at saturation (kJ/mol)	Average hydrogen bond length (Å)
AMO-mPD	273.053	2.54
AMO-EDOT	8.401	4.15
AMO-Py	63.013	1.86

of hydrogen bond(s) calculated for the interactions as well as the Mulliken charges of the different functional groups for each monomer-template interaction are shown in Fig. S1, S2 and S3 and Table S1, S2, S3. As it can be seen from Table 1, the total binding energy between mPD and AMO is significantly higher (273 kJ/mol) as compared to AMO interaction with other monomers, EDOT and Py. Thus, one could expect that the use of P(mPD) as a polymer matrix for AMO imprinting results in AMO-MIP with higher selectivity towards AMO as compared to the use of polypyrrole and PEDOT.

3.1.2. UV–vis and NMR study of prepolymerization complex formation

UV–vis spectroscopy was further used to characterize and establish the pre-polymerization complex formed between the monomer and the template molecules prior to the polymerization. For this purpose, the absorption spectra of the monomer, mPD and the template, AMO in the PBS buffer solution (pH = 7.4) were determined and compared to the spectra of their mixture in the concentration ratio 1:1. As shown in Fig. S4, the absorption spectra of mPD and AMO mixture shows a new but broad absorption band at around 284 nm that was different from either that of mPD (289 nm) or AMO (274 nm) as well as the disappearance of bands at 228 nm of AMO and 245 nm of mPD, indicating the presence of binding interactions between the molecules in the mixture. Thus, evidence of possible interaction between mPD and the templates were further established by UV–vis spectroscopic technique. Additionally, ^{13}C NMR spectra of the individual components as well as their mixture clearly indicate that there is no covalent interaction involved in the complex formation as no new peaks were observed in the AMO + mPD spectrum as compared to the individual spectra (Fig. S5). Therefore, it can be concluded that the complex formation is majorly through non-covalent interactions.

3.2. Electrodeposition of antibiotic-MIP films

The use of MIPs for sensing purposes generally demands a MIP-based material in the form of a uniform thin film and its robust interfacing with a sensor platform capable of responding to relevant sensitivity levels upon interaction between MIP film and a binding analyte. Electropolymerization has been shown to be especially suitable synthesis method for effectively interfacing MIP with transducer allowing a good control of both film thickness and inner morphology [16,23,24,28].

Therefore, the electrochemical deposition of P(mPD) was studied in detail by EQCM to provide a more accurate control of polymer film growth. To find optimal synthesis parameters for homogeneous polymer film preparation, the electropolymerization of mPD was studied by applying different modes of potential stimulus including, potentiostatic deposition (Fig. 3a), deposition by cyclic potential sweep (Fig. 3b) and potential pulse deposition (Fig. 3c). The P(mPD) films with thickness of 25 nm (as previously obtained from ellipsometry measurement) that correspond to the QCM frequency shift of 400 Hz (as calculated by using Eqs. (1) and (2)) were electrodeposited on the QCM sensor electrode. This relatively high thickness was selected so as to give a greater surface area of recognition for the small molecular weight target analyte. It was observed that in the case of polymer film deposition by potential cycling and potential pulses (Fig. 3b and c), the film growth rate was very low, reaching the 400 Hz frequency shift limit in not less than about 1 h. However, a P(mPD) film of the same thickness can be deposited for approximately 200 s by using the potentiostatic mode (Fig. 3a). This observation, coupled with the identified possible challenges of using potentiodynamic mode inclusive of disordered spatial chains of polymer matrix as well as film defects due to polymer swelling and shrinking, as compared to the well-ordered films produced by potentiostatic condition [16] suggest the preferred use of the latter for the polymer synthesis in this work. Potentiostatic synthesis at 0.6 V vs Ag/AgCl/KCl$_{sat}$ is then chosen to be the optimal parameter for preparing a well adherent and homogeneous P(mPD) films on the electrode sensor.

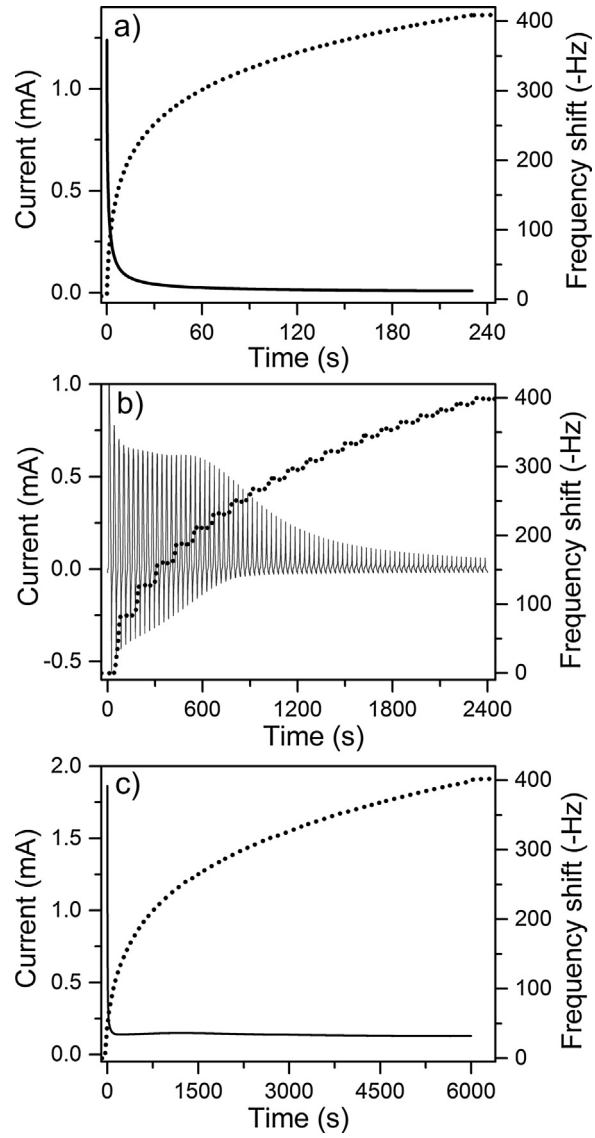

Fig. 3. Current responses during the electrodeposition of P(mPD) from PBS buffer solution, pH 7.4 under: (a) potentiostatic; (b) cyclic voltammetry; (c) potential pulse modes.

Furthermore, to exclude the possibility of the oxidation of the antibiotic template during the polymerization, an initial study of the electrochemical behavior of AMO in PBS buffer was carried out with the purpose of elucidating its electrochemical stability within the potential window used for the polymerization. For this purpose, cyclic voltammetry (CV) scan was applied to determine the

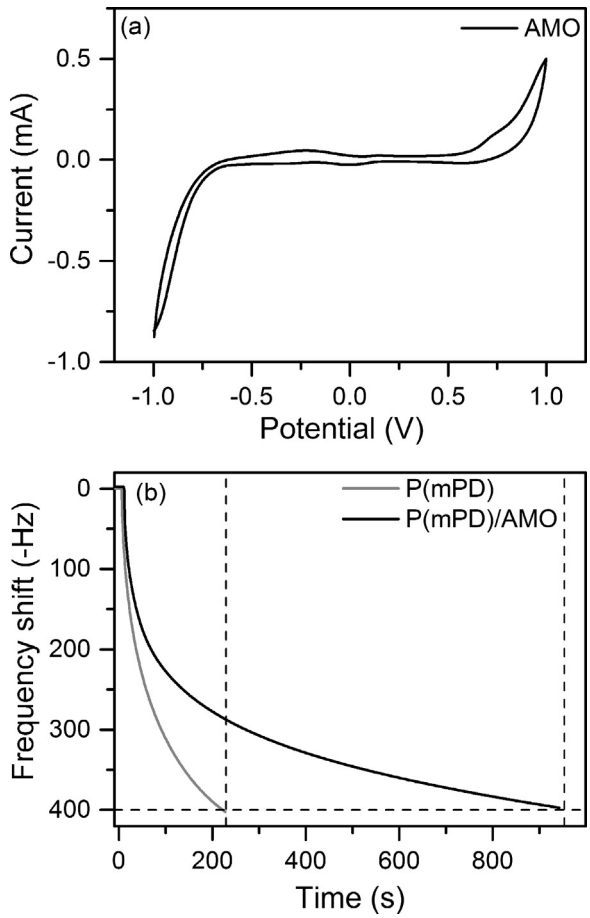

Fig. 4. (a) Cyclic voltammogram of a gold electrode in PBS buffers (pH 7.4) containing 5 mM AMO. Scan rate: 50 mV/s. (b) Resonant frequency responses during the potentiostatic electrodeposition of P(mPD) and P(mPD)/AMO films on gold electrode of QCM sensor.

electrochemical activity of the antibiotic within a specific potential window as shown in Fig. 4(a). The cyclic voltammogram indicate that the AMO molecules do not demonstrate any electroactivity at the potential (0.6 V) used for monomer polymerization hence, its stability during the polymer synthesis.

Fig. 4b shows typical resonance frequency responses of the EQCM during P(mPD) potentiostatic electrodeposition. As easily observed, the P(mPD) films grew non-linearly demonstrating a faster growth in the beginning of the process, followed by its gradual slowing-down until the constant resonant frequency value was reached, thus indicating a self-limiting polymerization. It should be noted that the film growth rate of the P(mPD) in the presence of the antibiotic was much slower than the rate for the pure P(mPD) film at the very same polymerization conditions. It is assumed that the dominant species of the targeted AMO antibiotic in PBS buffer solution (pH = 7.4) are negatively charged ions considering its acid-base properties [29]. Therefore, during the oxidative polymerization of mPD, negatively charged antibiotic species are expected to be inserted into the positively charged P(mPD) backbone due to electrostatic interactions, thus resulting in the formation of P(mPD) film containing the template molecules (P(mPD)/AMO).

3.3. Rebinding and selectivity study

For a rebinding of the target analyte on the P(mPD)/AMO film, the template molecules must be eluted from the polymer matrix to reveal complementary cavities in the film. This was achieved by treatment in solution mixture of acetic acid and methanol,

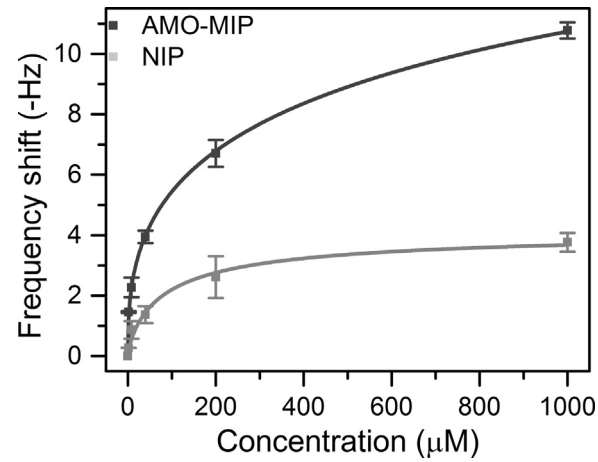

Fig. 5. Adsorption isotherms of the injection of increasing AMO concentration on AMO-MIP and NIP surfaces. Solid lines represent fits to the Langmuir–Freundlich model.

as discussed in the experimental (Section 2.3.2). However, the chromatographic analysis of the eluted solution indicates a low removal of the template molecules from the polymer matrix with an elution efficiency of about 20% thus indicating that some of the template molecules may still be trapped within the polymer. The analysis of the antibiotic rebinding capability of the prepared AMO-MIP film was achieved by employing the QCM-FIA technique. For a proper analysis of the rebinding interaction and to ascertain that the interaction between the target and the MIP film was specific, a control experiment was performed with the non-imprinted polymer (NIP) modified electrode sensor. The NIP was prepared like the MIP but excluding the template addition in the pre-polymerization mixture. To minimize signal fluctuation due to temperature change, the rebinding experiments were carried out in a closed custom-designed insulating chamber at room temperature. After equilibrating the film surfaces with the running buffer to obtain a stable baseline, the signal following injection of each concentration of analyte was monitored, recorded and analyzed by fitting to a pseudo first order kinetic equation to obtain the equilibrium responses shown in Fig. 5. These equilibrium signals data were fitted to the Freundlich-Langmuir (FL) model (Eq. (3)) using a nonlinear regression to obtain the kinetic data of the interaction. According to Shimidzu and co-workers [30], FL model is more universally applicable in characterizing MIPs because it can provide heterogeneity information and is able to model adsorption behavior over the entire range of concentration including both subsaturated and saturated region and therefore the FL model is most commonly used for characterizing MIPs [31].

As evident from Fig. 5 the MIP film clearly demonstrates the increased antibiotic adsorption compared to the NIP for all injections. Moreover, a very noticeable difference in the adsorption between the MIP and NIP was observed already after the injection of the antibiotic sample with the lowest concentration (1.6 μM), while the NIP film showed a low non-significant response. As the concentration increased, the adsorption became more pronounced for the MIP while the NIP film approaches saturation as the injected concentration becomes larger. The parameters obtained from the fitting of the binding isotherm to Eq. (3) are as shown in Table 2.

$$B = B_{\max} C^m / (K_D + C^m) \quad (3)$$

where C is the concentration of AMO in solution, B and $B_{\max}$ are the fractions of bound AMO and its saturation value, respectively, m is the heterogeneity index, which varies from 0 to 1, with increasing heterogeneity as m values decreases. K_D is the equilibrium dissoci-

Table 2
Parameters obtained from the LF fitting of the MIPs and NIPs binding isotherms.

	B_{max} (Hz)	K_D (μM)	m (Heterogeneity index)	IF (imprinting factor) B_{maxMIP}/B_{maxNIP}	R^2
AMO-MIP	−30.2 ± 2.4	28.8 ± 3.2	0.4 ± 0.02	7.2	0.998
NIP	−4.2 ± 0.4	36.5 ± 9.6	0.8 ± 0.06	–	0.969

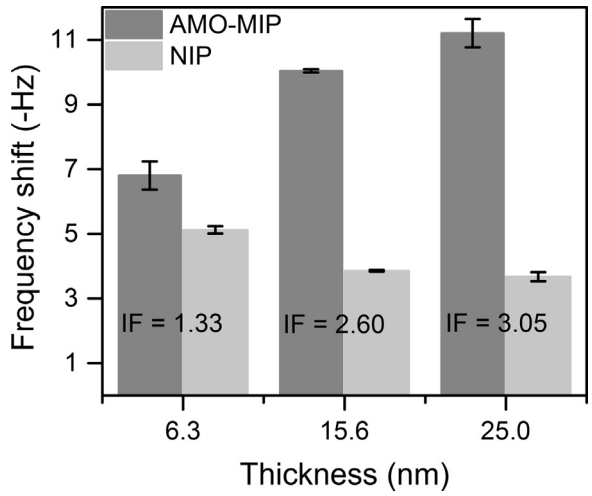

Fig. 6. Effect of film thickness on the analytical performance of AMO-MIP and NIP sensors at the injection of 1 mM AMO. Number labels on bars indicate the relative adsorption or imprinting factor (IF) values obtained at each thickness.

ation constant used to assess the affinity of the prepared MIP films to the template antibiotic.

From Table 2, one can observe that the AMO-MIP has a much higher adsorption capability as compared to the reference NIP film (-30.23 Hz vs −4.21 Hz) thus yielding an imprinting factor of 7.2. This indicate the relative capability of adsorption of the MIP compared to NIP. Likewise, the lower value of K_D for AMO-MIP relative to NIP indicates that the imprinted polymer has higher affinity for the target analyte as compared to the NIP, which mostly show non-specific interaction. This is because a lower K_D indicates a slower tendency of the target molecule to dissociate from the predefined active sites after rebinding and hence a strong binding affinity. This correlates well with the computational and spectroscopic studies that show a high binding energy, low hydrogen bond length and the presence of new UV–vis absorbance peaks in the solution mixture between AMO and mPD. Furthermore, since the removal of template molecules from the polymer matrix introduces within the MIP film binding sites with differing affinities towards the target, MIPs are generally expected to be more heterogeneous than their corresponding reference NIP films. Therefore, the lower m value observed for the AMO-MIP when compared to that of the NIP is quite familiar in MIP research. To characterize the effect of polymer matrix thickness on the analytical performance of the sensor, the AMO-MIP and NIP films with thicknesses of 6.3, 15.6 and 25.0 nm were interfaced with the sensor surface. This was realized by varying the time of electrodeposition process occurring directly on the sensor surface till the resonant frequency shift of the sensor reach −100, −250, and −400 Hz, respectively. Following the washing out process, the rebinding experiments were carried out with the injection of 1 mM concentration of AMO, as previously achieved (Fig. 6). As observed from the figure, an increasing thickness is accompanied by a corresponding increase in AMO-MIP adsorption and decrease in NIP adsorption. This is most likely due to the increase in specific surface area with increasing thickness and a corresponding increase in the number of binding sites in the AMO-MIP films. Consequently, an increasing adsorption is observed. In the case of NIP

Table 3
Parameters obtained from the fit of signals shown in Fig. 8.

Analyte	B_{max} (-Hz)	R^2
AMO	1.48 ± 0.03	0.999
DOXY	0.48 ± 0.03	0.999
SMZ	0.32 ± 0.01	0.997

however, the decreasing but stabilizing adsorption observed could be assumed to be associated with the generation of denser film layers with decreasing active surface area. This assumption stems from the nucleation and growth mechanism of polymerization similar to that of polyaniline, in which the density, uniformity and/or morphology of the final film depend on the interaction of nucleates with the underlying substrate. However, in the case of PmPD growth, the hydrophilic nucleates which adsorb on the hydrophobic polycrystalline gold substrate likely lead to the formation of rough profiles, island-like layer with pinholes that may permit access of template molecules to the electrode surface at low film thickness. But as polymer film grows, these pinholes are obstructed thus, decreasing the nonspecific adsorption signal hence, an increasing value of imprinting factor (IF) from 1.33 to 3.05 [32,33]. Furthermore, to test the selectivity of the prepared AMO-MIP, two non-targets but widely used antibacterials belonging to different antibiotics classes were selected and their binding responses on both AMO-MIP and NIP upon separate injections of increasing concentration were analyzed and compared to that of AMO. Fig. 7a indicates the ratio of the equilibrium binding response of each analyte on the AMO-MIP to that on the NIP. As expected, the imprinted surface had a greater relative adsorption for amoxicillin target molecules. In contrast, the adsorption of the non-template antibiotics, SMZ and DOXY, was revealed to be considerably lower. Considering the little higher molecular weight of DOXY (444.4 g/mol) as compared to AMO (365.4 g/mol), one may expect DOXY's response signal on AMO-MIP to be quite comparable with that of AMO, especially at low concentration but a quick look at Fig. 7a indicates the opposite. This thus prompts the closer attention into the signals received from the NIP film following injection of the different concentrations of all samples. As shown in Fig. 7b, a greater non-specific adsorption is displayed by the NIP surface towards the non-targeted molecules as compared to AMO. This observation leads to the reasonable conclusion that the major part of the adsorption response displayed on the AMO-MIP by SMZ and DOXY are non-specific in nature and different from the specific recognition of the target AMO. Furthermore, the signal obtained from the injection of 1.6 μM AMO following a prior injection of 1.6 μM SMZ and DOXY (Fig. 8) indicates that the prior injection of other antibiotics does not influence, to any noticeable extent, the selective recognition of AMO on AMO-MIP.

Thus, the AMO adsorption equilibrium response as shown in Table 3, reflects at least 3 times higher signal with respect to both SMZ and DOXY at this concentration. More importantly, the selectivity coefficients (α) of the MIP film, defined by the index of the polymer selectivity towards other analytes, was calculated (Table 4) according to Eq. (4) for increasing concentration of each analyte taking cognizance of their molecular weight.

$$\alpha = (B_{max(a)}/M_{(a)})/B_{max(t)}/M_{(t)}) \quad (4)$$

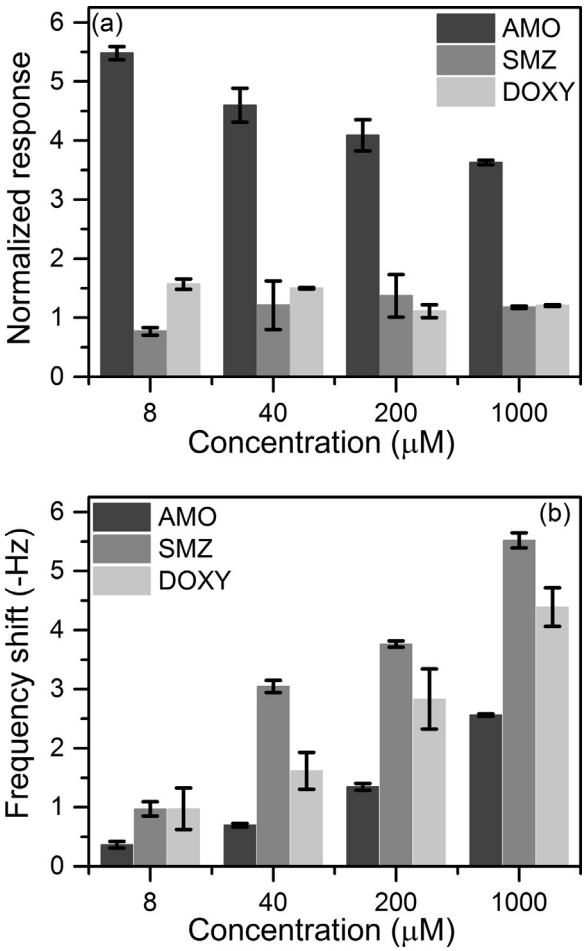

Fig. 7. (a) Normalized equilibrium resonant frequency responses of AMO-MIP modified QCM sensors upon injection of different antibiotics. (These data were obtained by dividing the response of the AMO-MIP by that of the NIP for each analyte injection) (b) Equilibrium resonance frequency responses of NIP film after injection of increasing concentration of the three different antibiotics.

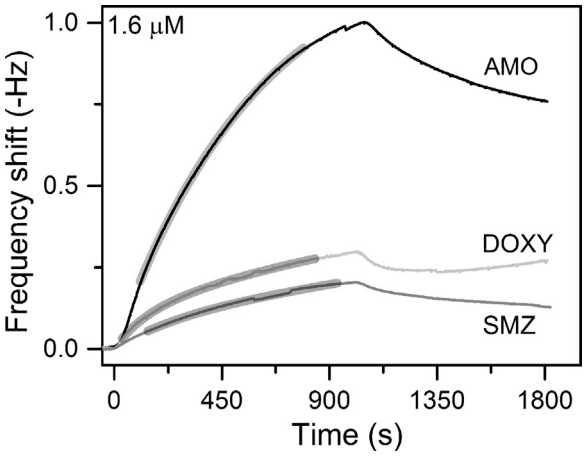

Fig. 8. Typical response signals of the injection of 1.6 μM concentration of each of the sample solution on AMO-MIP film sensor. Thick lines represent fits to the pseudo first-order equation.

where $B_{max(a)}$ and $B_{max(t)}$ represent responses of AMO-MIP (in Hz) toward non-imprinted analyte and template molecules possessing molecular weights of $M_{(a)}$ and $M_{(t)}$, respectively.

The low values of the selectivity coefficients as shown in Table 4 indicates that the polymer selectively recognizes the template molecules but non-specifically, and with very low affinity (similar to the non-imprinted polymers) binds non-templated interfering molecules. We also went further to analyze the lowest detectable concentration of the analyte by the prepared QCM sensor. For this purpose, the limit of detection (LoD) value was calculated from the standard deviation and slope of the regression line for the linear portions of the response signals at low concentrations (2–40 nM) injections (Fig. S6). A value of 0.2 nM was obtained, indicating that the sensor can detect the target analyte down to this low concentration. Since AMO exists in the aqueous environment within a concentration range of 0.18–4.57 nM as earlier reported, the developed chemosensor shows very promising suitability for detecting the target molecule within the low concentration range that they exist naturally. Thus, the prepared AMO-MIP sensor demonstrates an immense potential for selectively recognizing AMO antibiotic in aqueous media. Furthermore, to characterize the sensor for reusability and stability, we perform several cycles of regeneration-rebinding experiments on both AMO-MIP and NIP sensors (Fig. 9). It was observed that after the third cycle, AMO-MIP and NIP films have lost 37% and 49% adsorption capacities respectively. This result may likely reveal that with subsequent regeneration process, less bound analytes are eluted from the polymer film thus reducing the chances of analyte binding in the following rebinding step. The results therefore demonstrate that the AMO-MIP is quite stable and reusable for at least three cycles while still retaining almost 70% of its adsorption capacity.

Table 4
Response-Selectivity Coefficients (α) of AMO-MIP Calculated Using Eq. (4) for Sample Solutions Containing Different Concentrations of Antibiotics, SMZ and DOXY.

Concentration (μM)	SMZ	DOXY
1.6	0.12	0.34
8.0	0.39	0.49
40.0	0.48	0.42
200.0	0.54	0.32
1000.0	0.56	0.35

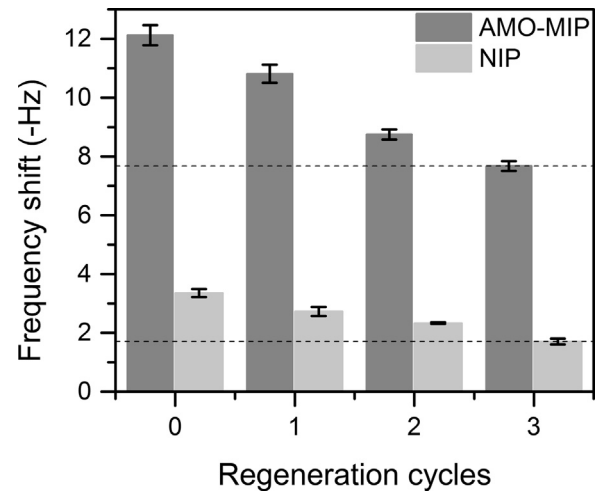

Fig. 9. Effects of regeneration-rebinding cycles on the adsorption capacity of both AMO-MIP and NIP films at the injection of 1 mM AMO.

4. Conclusions

This study draws special attention to the development of a selective MIP for amoxicillin detection by synthesizing homogeneous thin polymer film on QCM sensor using electrochemically polymerizable m-phenylenediamine (mPD) as a functional monomer. A careful study of the electrodeposition of the polymer in the

presence or absence of the template molecules was carried out to ascertain the optimal parameters for the fast synthesis of a homogeneous and stable polymer films that affords the template stability during the polymer electrodeposition. The prepared AMO-MIP film as well as its reference counterpart were characterized by QCM technique for their affinity and selectivity towards the target analyte. The AMO-MIP film showed at least seven times higher binding capacity and better affinity than the reference film. The prepared AMO-MIP modified QCM sensor could detect AMO with LoD of 0.2 nM and discriminate between the antibiotics of different classes; i.e. tetracycline (doxycycline) and sulfonamide (sulfamethizole). Further research is being undertaken to optimize the performance of the sensor and to enhance its capacity in detecting amoxicillin in real aqueous environment.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.snb.2017.11.194.

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